PI Protocol: Fullstack Privacy

Logic Chaining & Feature Implementation

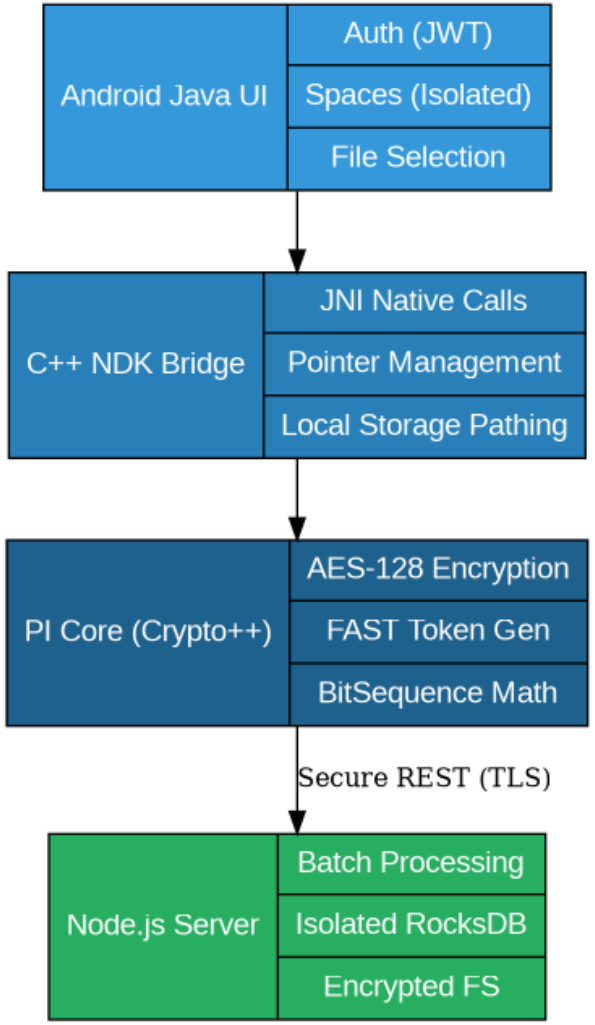
Deep Dive Overview

March 2026

The Logic Chain: Data Journey

How data moves safely:

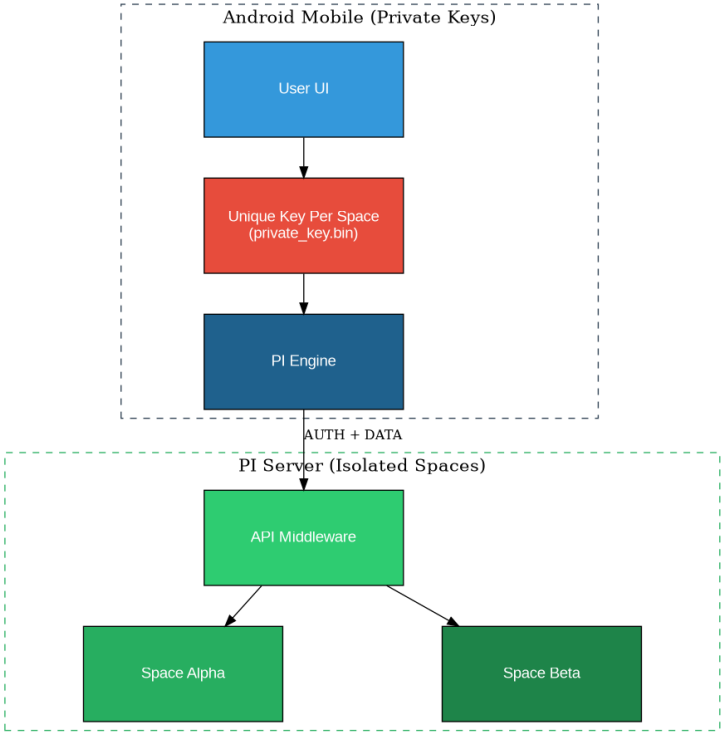
- **Selection**: Java layer selects files/keywords.
- **Processing**: NDK invokes C++ PI Engine.
- **Tokenization**: FAST scheme generates 200k tokens.
- **Batched Sync**: Tokens sent in chunks of 5,000.



Isolated Secure Spaces

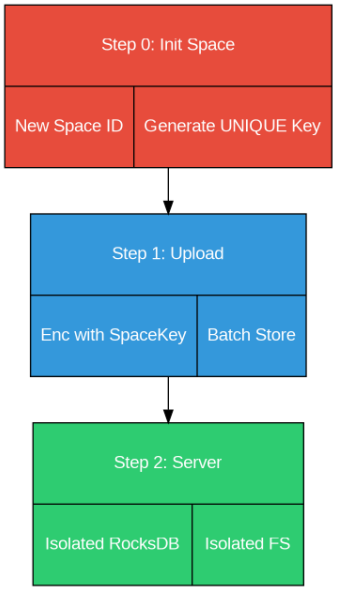
Zero-Trust Isolation:

- **Physical Separation**: Every space has a unique RocksDB + FS folder.
- **Unique Keys**: Private keys generated per-space; never leave the phone.
- **Session Security**: JWT authentication protects all space interactions.

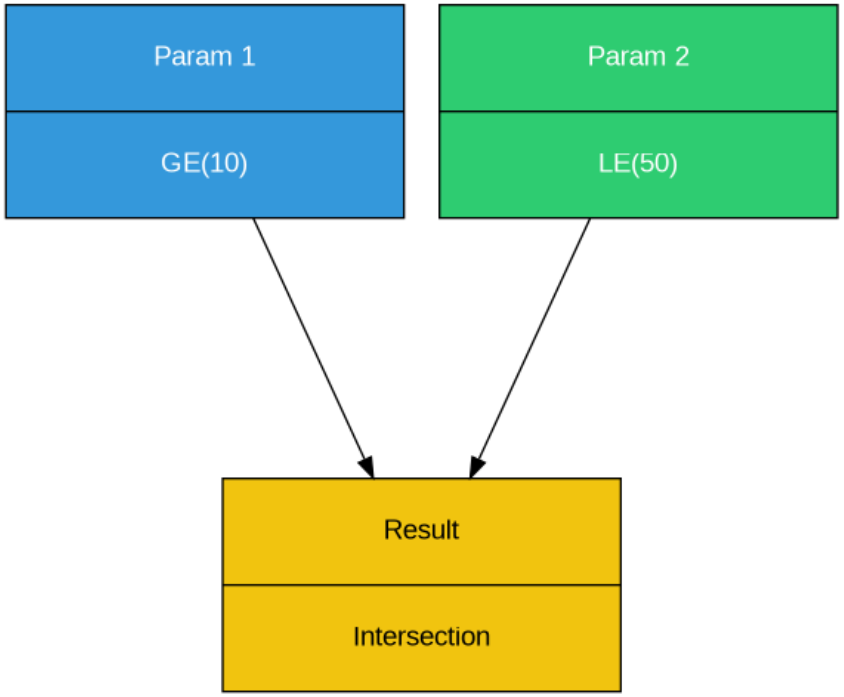


Advanced Update & Metadata

- **Extension Preservation**: Encrypts `doc.pdf` to `ID0.pdf`. The server knows the extension (so storage works) but not the content.
- **Payload Batching**: The App prevents Out-Of-Memory (OOM) network crashes by chunking uploads into exactly 5,000-token arrays and 5-file bundles per HTTP POST.
- **Local State Management**: The phone tracks indexing state locally in a mini-RocksDB to prevent token collision.



Private Search Strategy



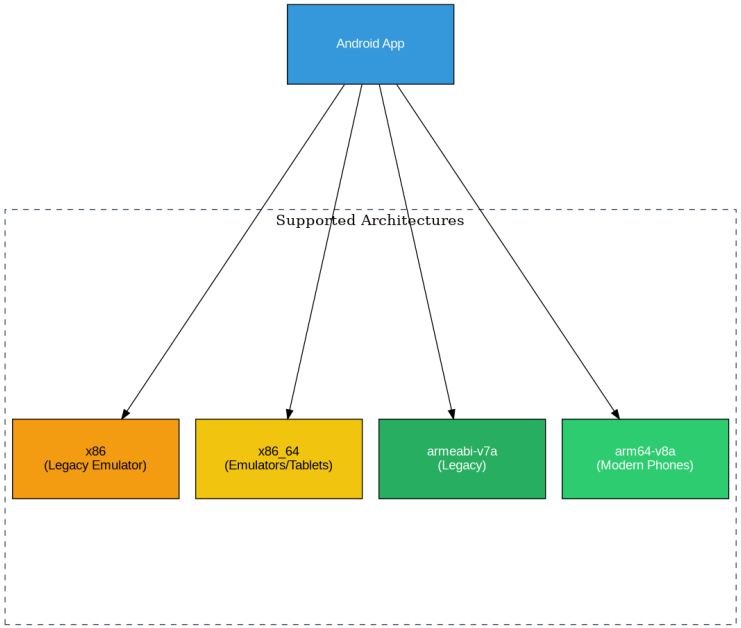
How Range Queries Work:

- **Trapdoors**: App generates 2 secure search tokens.
- **Server Side**: Returns encrypted match sets.
- **Intersection**: `BitSequence` logic on phone finds the final result.

Universal Android Support

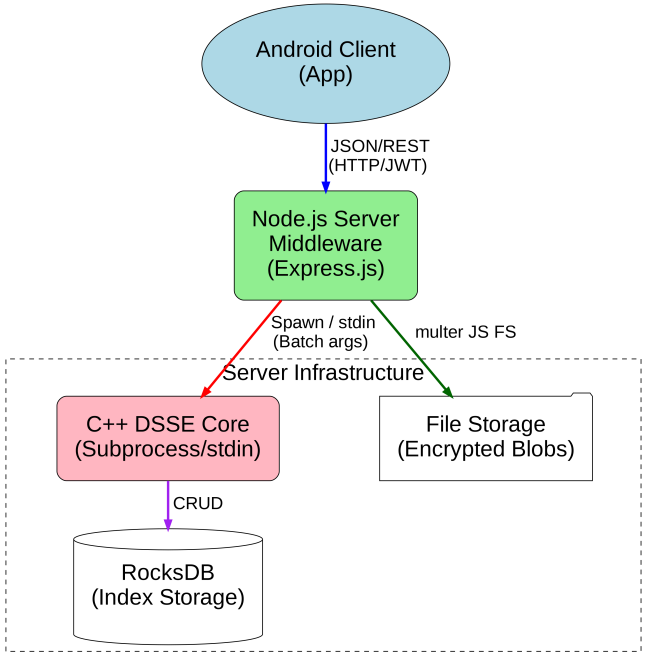
Multi-Core Stability:

- Supports **arm64-v8a**, **v7a**, **x86**, and **x64**.
- High-speed static C++ libraries linked via CMake.
- Hardened JNI to prevent native crashes.



Server Architecture: Node.js Middleware

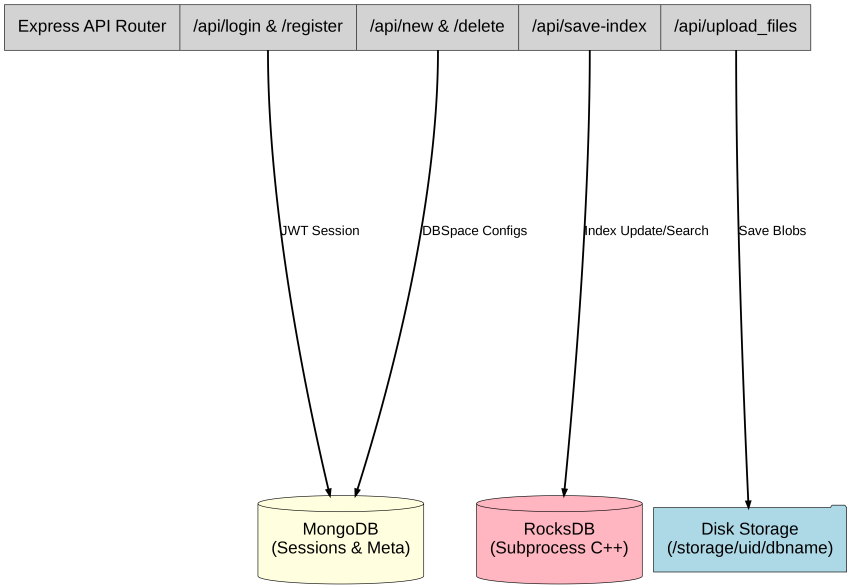
- **Role**: Acts as a secure REST middleware bridging the Android client with the C++ DSSE Core and File Storage.
- **Decoupling**: Abstracts away complex command-line native executions (RocksDB) into predictable, standard REST calls.
- **Native Integration**: Spawns `dsse_server` child processes, feeding large batch updates via `stdin` streams to bypass OS command-line limits.



API Endpoints & Security

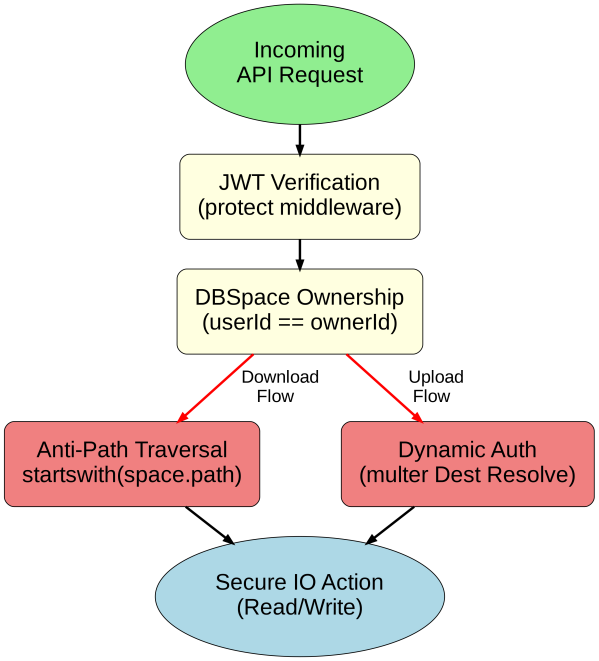
Endpoint Structure & Purpose:

- **Auth** (`/login`): JWT-based sessions in MongoDB.
- **Space** (`/new`): Provisions isolated `DBSpace`.
- **Index** (`/bulk-save`): Interacts with RocksDB for search/update.
- **File** (`/upload`): Handles encrypted document blobs.



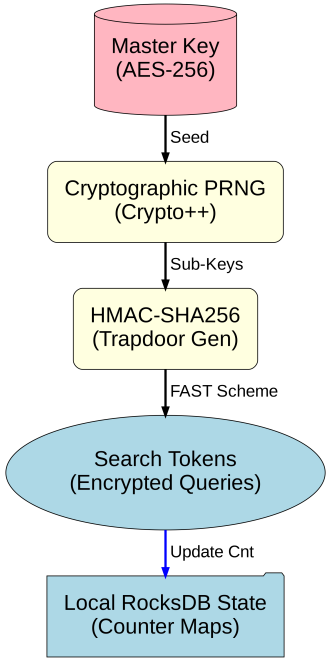
Security & Edge Case Handling

- **Zero-Trust**: `protect` middleware validates JWT AND ownership.
- **Immutable Locks**: Upon completion, the App triggers `/lock-space`. The API irrevocably sets `isLocked=true` in Mongo, physically preventing hackers from re-uploading and destroying the RocksDB index.
- **Anti-Path Traversal**: Verifies `startsWith()` against `../` attacks.
- **Dynamic Storage**: `multer` safely resolves path before bytes read.



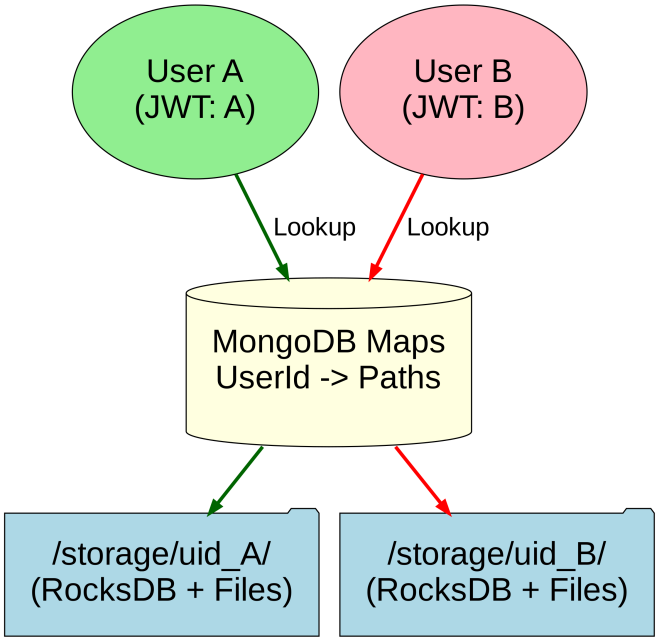
Cryptographic Core: The FAST Scheme

- **C++ & NDK**: High performance native cryptographic routines using `Crypto++`.
- **Master Key Derivation**: Initial AES-256 keys generate subsequent sub-keys via Pseudo-Random Number Generators (PRNG).
- **HMAC-SHA256**: Encrypts both the file paths and search keywords into Trapdoor Tokens.
- **Local Counters**: Phone retains state integers via a local RocksDB preventing duplicate tokens.

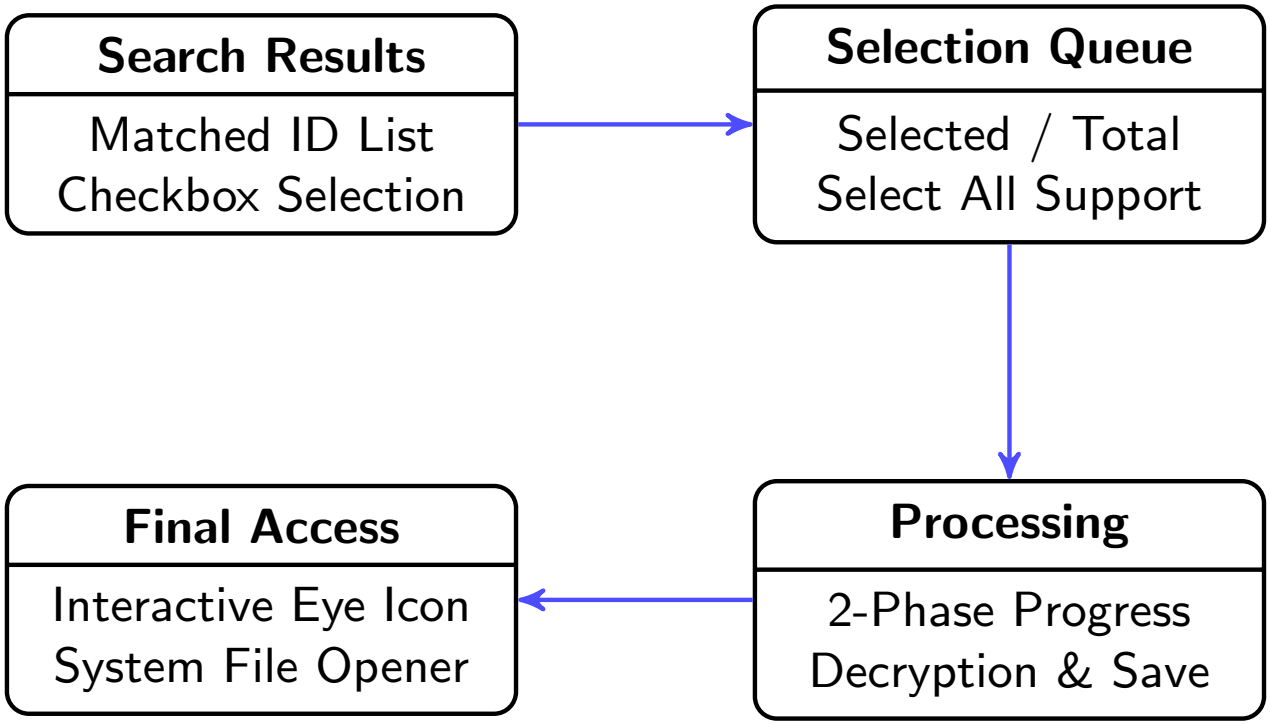


Multi-Tenant Storage Isolation

- **Complete Compartmentalization**: Each user's database is a 100% physically separate folder on the Linux server filesystem.
- **JWT → UID Mapping**: The API extracts the authenticated UID from the Bearer Token, never trusting the client's payload.
- **MongoDB Lookup**: The DBSpace schema strictly links the requested database name to the verified UID to extract the true absolute path (`/storage/uid_XYZ/dbname/`).



Granular Download Control: Logic Flow



- **Efficiency**: Targeted downloads reduce server load.
- **UX**: Instant visual feedback with green success states.

UX Architecture: Live Monitoring

- **Luxury Visuals**: Midnight Surface with Luxury Gold branding.
- **2-Phase Interactive Progress**:
  - **Phase 1 (Gold)**: Real-time network stream tracking (0–80%).
  - **Phase 2 (Green)**: Post-decryption success verification (100%).
- **Adaptability**: Full `fitsSystemWindows` support for hardware bezels.
- **Stable Scrolling**: Nested scroll containers prevent layout shifting with dense data.

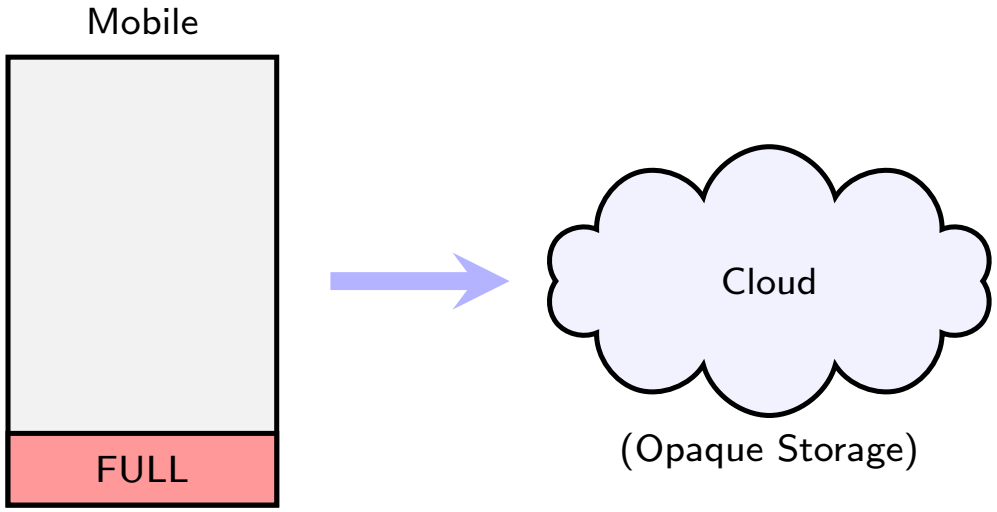
App Functionality Summary

1. **Login**: Secure JWT session establishment.
2. **Update**: Batch token generation & file encryption.
3. **Search**: Range query with **Manual Selection Control**.
4. **Live Recovery**: 2-phase tracking with **Integrated Viewing**.

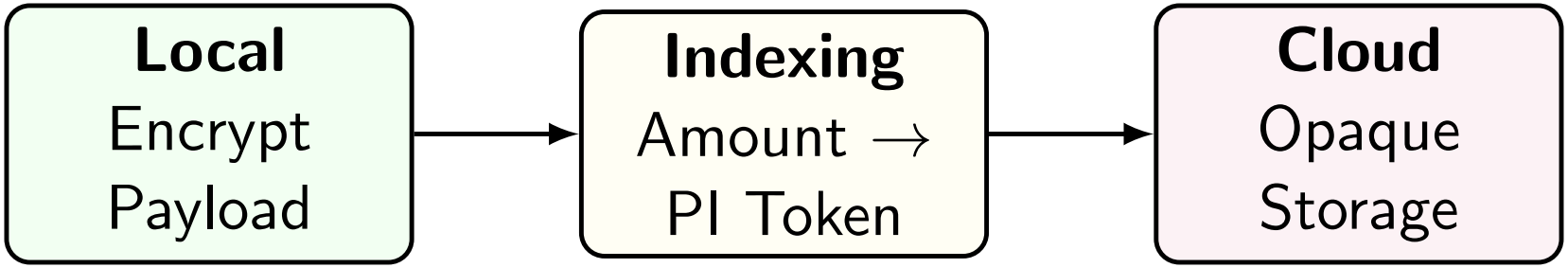
DEMO: The Storage Challenge

Problem Statement:

- **Data**: Tax Defaulter PDFs
- **Issue**: Phone storage full
- **Risk**: Untrusted Cloud



DEMO: The PI Searchable Workflow



- **Zero Visibility**: Server only stores bit-pattern indexes.
- **Selective Control**: Phone requests only what it needs.

DEMO: Tax Defaulter Dataset Sample

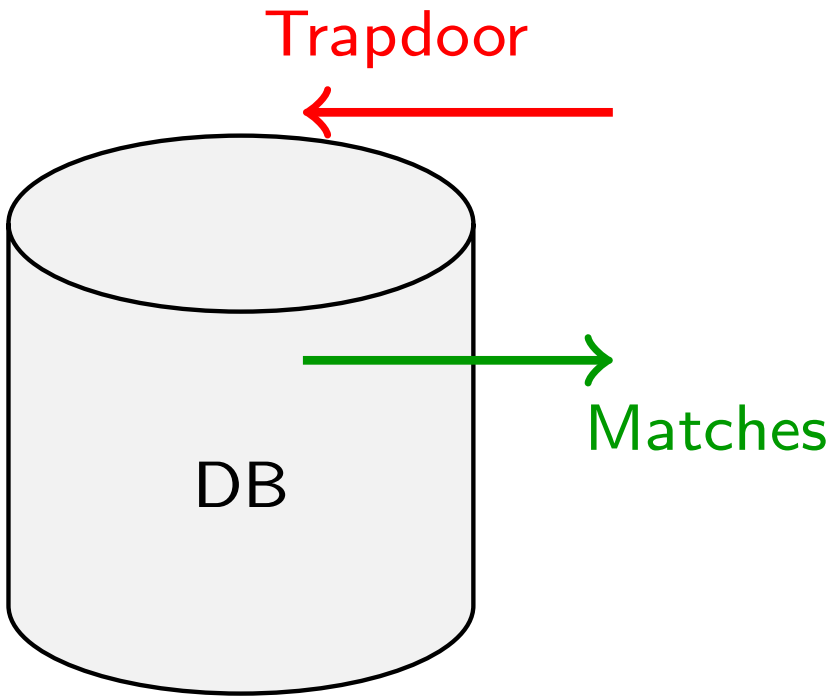
File (Keyword)	Taxpayer Name	Status
99286.pdf	Sofia Oliveira	Defaulter
74127.pdf	Li Wei	Defaulter
65386.pdf	Lucas Muller	Defaulter
58625.pdf	John Smith	Defaulter
14747.pdf	Maria Garcia	Defaulter

Amounts (14k–99k) serve as secure range-search keywords.

DEMO: Range Query [$50k – $75k]

In-App Query:

- **Input**: Range [50k, 75k]
- **Trapdoor**: Client secret
- **Retrieve**: 3 Files



Outcome: Only relevant files retrieved via Range Query.